Lecture 13: Characteristics vs Covariances

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Intro

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- Last lecture gave us traded factors: market, SMB, HML, and momentum (Fama and French 1993; Carhart 1997)
- Today: what do those factors actually represent?
- Risk exposures?
- Or firm characteristics dressed up as factors?

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- Today: what do those factors actually represent?
- Risk exposures?
- Or firm characteristics dressed up as factors?
- What if betas are moving around and we are mismeasuring them?
- Building blocks for a modern view of this debate (Kelly, Pruitt, and Su 2019)

Flight Plan

- Part I: what is being priced? (Fama and French 1993; Daniel and Titman 1997)
- Part II: the empire strikes back (Davis, Fama, and French 2000)
- Part III: what if betas are moving around? (Ferson and Harvey 1999)
- Part IV: pricing the 25 portfolios is easy like Sunday morning (Lewellen, Nagel, and Shanken 2010)
- Part V (next class): a modern reconciliation (Kelly, Pruitt, and Su 2019)

Part I: What Is Being Priced?

Two Competing Stories

- Fama and French (1993) showed that SMB and HML help price test portfolios
- But pricing success does not pin down economic interpretation
- A factor can work in the data for more than one reason

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- But pricing success does not pin down economic interpretation
- A factor can work in the data for more than one reason
- Daniel and Titman (1997) challenged the risk-based reading of SMB and HML
- Story 1 (Fama and French 1993): expected returns are driven by factor loadings, the betas generate premia
- Story 2 (Daniel and Titman 1997): expected returns are driven by characteristics directly

Mathematical Interpretation

Fama and French (1993) propose that:

$$\mathbb{E}[R_{i,t} - R_{f,t}] = \beta_i^\top \lambda$$

where λ is a vector of factor premia.

- Important: the only "charachteristics" that matter are the betas!

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Daniel and Titman (1997) propose that:

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where z_i is a vector of characteristics (size, leverage, profitability, book-to-market ratio, ...) and g is an unknown function.

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- What Ross (1976) would say about this discussion?

Let’s Watch a Video

Click here

Time: 2:00–6:50

Let’s try to reflect on the manager’s idea through these two different lenses!

An Ideal Test

- Imagine we could see two firms with the *same* factor loadings but *different* characteristics
- Under Fama and French (1993), they should have the same expected return
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An Ideal Test

- Imagine we could see two firms with the *same* factor loadings but *different* characteristics
- Under Fama and French (1993), they should have the same expected return
- Under Daniel and Titman (1997), they *could* have different expected returns
- Problem: empirically, this is essentially impossible to find
- Characteristics and factor loadings are highly correlated in the data – almost by construction
- DT97's key contribution: design a test that tries to break this link

Daniel and Titman: The Design

- Construct 9 portfolios by sorting stocks on size and book-to-market (3x3)
- Each June at time t, estimate the 3-factor model for each stock using past data
- For each of the 9 portfolios, create 5 sub-portfolios based on the estimated β_{HML}
- Total: 45 portfolios sorted on (size, book-to-market, and HML beta)

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- Total: 45 portfolios sorted on (size, book-to-market, and HML beta)
- Under Story 2 (characteristics), the sub-portfolios should have similar returns
- Under Story 1 (covariances), the sub-portfolios should have different returns that track their HML betas

Characteristic-Matched Portfolios

- Let $R_{s,b,h}$ denote the return of the portfolio with size sort s, book-to-market sort b, and HML beta sort h
- $h = 1$ is the lowest HML beta, $h = 5$ is the highest

Characteristic-Matched Portfolios

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- $h = 1$ is the lowest HML beta, $h = 5$ is the highest
- Define the "characteristic-matched" portfolio return $C_{s,b}$ as

$$C_{s,b} \equiv \underbrace{R_{s,b,1} + R_{s,b,2}}_{\text{low HML beta}} - \underbrace{R_{s,b,4} - R_{s,b,5}}_{\text{high HML beta}}$$

- There are 9 of these portfolios, one for each (size, book-to-market) cell

The Test

- Consider estimating

$$C_{s,b} = \alpha_{s,b} + \beta_{s,b,Mkt} \cdot MKT_t + \beta_{s,b,SMB} \cdot SMB_t + \beta_{s,b,HML} \cdot HML_t + \varepsilon_{s,b}$$

- If FF93 is right, we should find $\alpha_{s,b} \approx 0$
- They show that SMB and MKT loadings are similar across $R_{s,b,h}$
- So we would expect $\mathbb{E}[C_{s,b}] < 0$ under FF93

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But if DT97 is right...

- We should find $\alpha_{s,b} > 0$ since $ _{s,b} = $ (true premium) - (what FF93 implies)
- We should also find $\mathbb{E}[C_{s,b}] \approx 0$

Empirical Results - MKT Betas

Char Port		$\widehat{\beta}_{\mathrm{Mkt}}$				
BM	SZ	1	2	3	4	5
1	1	1.12	1.03	1.07	1.04	1.15
1	2	1.14	1.03	1.03	1.07	1.08
1	3	0.99	0.98	0.95	1.04	1.04
2	1	0.99	0.93	0.95	0.95	1.08
2	2	1.06	0.96	0.94	1.01	1.06
2	3	0.97	1.02	0.96	1.04	1.07
3	1	1.01	0.92	0.94	1.05	1.17
3	2	1.05	0.99	0.98	1.02	1.20
3	3	1.02	1.03	0.99	1.03	1.15
Average		1.04	0.99	0.98	1.03	1.11

Empirical Results - HML and SMB Betas

Char Port		$\hat{\beta}_{\text{HML}}$					$\hat{\beta}_{\text{SMB}}$				
BM	SZ	1	2	3	4	5	1	2	3	4	5
1	1	−0.40	−0.38	−0.11	−0.04	0.25	1.23	1.07	1.07	1.18	1.39
1	2	−0.60	−0.32	−0.18	−0.05	0.05	0.81	0.55	0.62	0.55	0.61
1	3	−0.70	−0.44	−0.22	−0.11	−0.02	−0.14	−0.17	−0.16	−0.08	0.04
2	1	0.02	0.19	0.32	0.35	0.48	1.19	0.95	0.94	0.89	1.15
2	2	0.17	0.23	0.28	0.36	0.49	0.54	0.45	0.44	0.47	0.72
2	3	0.03	0.24	0.22	0.31	0.49	−0.22	−0.25	−0.15	−0.10	−0.07
3	1	0.42	0.50	0.57	0.75	0.91	1.24	1.01	0.95	1.04	1.25
3	2	0.40	0.58	0.56	0.72	0.82	0.61	0.43	0.37	0.43	0.69
3	3	0.45	0.56	0.67	0.81	1.00	−0.06	−0.15	−0.17	0.05	0.10
Average		−0.02	0.13	0.23	0.34	0.50	0.58	0.43	0.43	0.49	0.65

Empirical Results - Main Table

Char Port		Char-Balanced Portfolio: t-Statistics				
BM	SZ	$\hat{\alpha}$	β_{Mkt}	β_{SMB}	β_{HML}	R^2
1	1	1.43	−0.43	−2.69	−9.21	31.48
1	2	0.50	0.18	1.98	−8.99	31.48
1	3	−0.48	−1.62	−2.52	−8.57	27.11
2	1	1.37	−2.02	1.31	−7.13	18.43
2	2	2.12	−0.99	−2.07	−4.69	10.96
2	3	0.79	−1.41	−2.34	−3.96	9.11
3	1	2.53	−5.30	−0.48	−8.00	23.36
3	2	2.01	−2.30	−0.63	−4.52	8.58
3	3	1.08	−1.30	−2.36	−4.98	12.39
Combined portfolio		0.354	−0.110	−0.134	−0.724	41.61
Expected return: -0.116%		(2.30)	(−3.10)	(−2.40)	(−12.31)	

Part II: Follow-Ups

Davis-Fama-French: The Reply

- Davis, Fama, and French (2000) push back on the characteristic story
- Their point: the sample in DT97 (1963-1993) is short and anomalous
- They perform the same analysis as DT97 but with different samples, getting different results

Period	Ave	t(Ave)	a	b	s	h	$t(a)$	$t(b)$	$t(s)$	$t(h)$	R^2
7/29–6/97	0.12	1.56	−0.06	−0.01	0.03	0.38	−0.83	−0.48	0.91	11.92	0.29
7/29–6/63	0.19	1.49	0.01	−0.01	0.06	0.35	0.11	−0.19	1.09	6.99	0.24
7/63–6/97	0.05	0.56	−0.14	0.01	−0.01	0.43	−2.07	0.48	−0.28	14.32	0.42
7/73–12/93	0.03	0.25	−0.22	0.02	0.03	0.46	−2.28	0.61	0.80	11.35	0.44
7/29–6/72 & 1/94–6/97	0.16	1.63	−0.00	−0.01	0.03	0.36	−0.01	−0.31	0.74	8.80	0.26

The Fight Kept On...

- Daniel, Titman, and Wei (2001) showed support for Daniel and Titman (1997) using data from the Japanese stock market
- Fama and French (1998) also showed support for a value factor (HML) in several international markets
- Fama and French (2012) showed that the size, value, and momentum factors are necessary to price stocks across many markets
- Important: localization matters! The markets might not be so integrated...

Questions?

Part III: What If Betas Are Moving Around?

What if Betas Are Not Constant?

- Everything we talked so far was about **unconditional** betas and premia
- But why should these things be constant over time?
- Companies change over time, so their betas should change too
- Different factors probably are born and die... no AI 15 years ago! Also, climate risk?
- Ferson and Harvey (1999) asks: what if we are mismeasuring betas by assuming they are constant?
- Can CAPM be saved? Are Fama French betas moving around?

Conditional vs Unconditional Models

- Unconditional model: $\mathbb{E}[R_{i,t} - R_{f,t}] = \beta_i^\top \lambda$
- Conditional model: $\mathbb{E}_t[R_{i,t+1} - R_{f,t+1}] = \beta_{i,t}^\top \lambda_t$

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The conditional model in general *does not imply* the unconditional one:

$$\mathbb{E}[R_{i,t} - R_{f,t}] = \mathbb{E}[\beta_{i,t}^\top \lambda_t] = \mathbb{E}[\beta_{i,t}]^\top \mathbb{E}[\lambda_t] + \mathsf{Cov}(\beta_{i,t}, \lambda_t)$$

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Can you think of examples in which $\mathsf{Cov}(\beta_{i,t}, \lambda_t) > 0$?

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More importantly: we might be measuring alphas wrong!

A Model With Time-Varying Betas

Assume we have K factors and asset i has loading $\beta_{i,t}$:

$$r_{i,t} = \beta_{i,t}^\top f_t + \varepsilon_{i,t}$$

where $r_{i,t}$ is the excess return of asset i at time t.

- Assumptions: $\mathbb{E}_t[\varepsilon_{i,t+1}] = 0$ and $\mathbb{E}_t[\varepsilon_{i,t+1} f_{t+1}] = 0, \forall i, t$

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- Assumptions: $\mathbb{E}_t[\varepsilon_{i,t+1}] = 0$ and $\mathbb{E}_t[\varepsilon_{i,t+1} f_{t+1}] = 0, \forall i, t$
- Let's generalize the expect return:

$$\mathbb{E}_t[r_{i,t+1}] = \alpha_{i,t} + \beta_{i,t}^{\top} \lambda_t$$
$$\alpha_{i,t} = a_{0,i} + a_{1,i}^{\top} z_{t-1}$$
$$\beta_{i,t} = b_{0,i} + b_{1,i}^{\top} z_{t-1}$$

where z_t is a vector of **instruments**

How Does Fama-French Fit Here?

- The instruments z_t should ideally be things known to forecast returns
- Question: do they help forecasting returns through betas? Or through $\alpha_{i,t}$?
- If the Fama-French model is conditionally correct, $a_{0,i} = 0$ and $a_{1,i} = 0$

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- If the Fama-French model is conditionally correct, $a_{0,i} = 0$ and $a_{1,i} = 0$
- How well can this model price assets?
- Can we use the Fama-French model as a conditional model?
- Do we have evidence that betas are changing over time in a systematic way?

Clever Estimation

Ferson and Harvey (1999) proposes a neat way of estimating these parameters:

$$r_{i,t} = \alpha_{0,i} + \alpha_{1,i}^\top z_{t-1} + (b_{0,i} + b_{1,i}^\top z_{t-1})^\top f_t + \varepsilon_{i,t}$$

- Regress realized excess returns on constants, instruments, and interactions between factors and instruments
- f_t: Fama-French factors
- $r_{i,t}$: 25 size-B/M portfolios
- z_t: 5 instruments (dividend yield, term spread, default spread, T-bill rate, and 3-1 yield spread)

Are Betas Time-Varying? Testing $b_{1,i} = 0$

	Time-Varying Alphas			Constant Alphas		
Portfolio	R^2 Constant Betas	R^2 Time-Varying Betas	F-test (p-value)	R^2 Constant Betas	R^2 Time-Varying Betas	F-test (p-value)
S1/B1	0.673	0.685	0.002	0.651	0.659	0.014
S1/B2	0.693	0.703	0.004	0.681	0.689	0.020
S1/B3	0.688	0.701	0.001	0.673	0.682	0.012
S1/B4	0.647	0.663	0.001	0.633	0.645	0.007
S1/B5	0.608	0.624	0.002	0.592	0.604	0.008
S2/B1	0.783	0.787	0.037	0.774	0.777	0.125
S2/B2	0.786	0.795	0.002	0.775	0.779	0.047
S2/B3	0.758	0.769	0.001	0.756	0.763	0.009
S2/B4	0.765	0.775	0.001	0.758	0.764	0.019
S2/B5	0.706	0.721	0.000	0.702	0.711	0.007
S3/B1	0.835	0.838	0.040	0.832	0.834	0.107
S3/B2	0.845	0.850	0.006	0.838	0.839	0.210
S3/B3	0.803	0.807	0.026	0.800	0.801	0.147
S3/B4	0.795	0.800	0.018	0.791	0.794	0.057
S3/B5	0.730	0.737	0.013	0.729	0.733	0.069
S4/B1	0.879	0.878	0.730	0.878	0.877	0.736
S4/B2	0.900	0.904	0.001	0.898	0.901	0.004
S4/B3	0.861	0.862	0.126	0.859	0.859	0.272
S4/B4	0.785	0.787	0.199	0.786	0.788	0.152
S4/B5	0.751	0.761	0.002	0.751	0.761	0.003
S5/B1	0.878	0.881	0.024	0.877	0.878	0.179
S5/B2	0.911	0.913	0.010	0.911	0.914	0.008
S5/B3	0.832	0.837	0.009	0.831	0.836	0.012
S5/B4	0.772	0.773	0.356	0.774	0.775	0.231
S5/B5	0.643	0.648	0.067	0.644	0.649	0.062

Are Alphas Time-Varying? Testing $a_{1,i} = 0$

Portfolio	Annual Intercept (Constant alpha, constant betas)	Test Zero Unconditional Alpha	Test Constant Alpha (Constant betas)	Test Constant Alpha (Time-varying betas)
S1/B1	-6.036	0.000	0.000	0.000
S1/B2	-1.924	0.036	0.002	0.002
S1/B3	-0.880	0.237	0.000	0.000
S1/B4	0.585	0.425	0.000	0.000
S1/B5	0.170	0.815	0.000	0.000
S2/B1	-0.917	0.320	0.002	0.002
S2/B2	-0.465	0.551	0.000	0.001
S2/B3	0.893	0.274	0.001	0.001
S2/B4	0.723	0.303	0.042	0.050
S2/B5	0.034	0.966	0.000	0.000
S3/B1	-1.100	0.239	0.000	0.000
S3/B2	0.100	0.908	0.002	0.003
S3/B3	-0.347	0.683	0.002	0.003
S3/B4	0.672	0.408	0.003	0.004
S3/B5	0.960	0.294	0.001	0.001
S4/B1	1.324	0.154	0.004	0.005
S4/B2	-2.322	0.011	0.000	0.000
S4/B3	-0.963	0.310	0.000	0.000
S4/B4	-0.040	0.969	0.000	0.000
S4/B5	0.476	0.693	0.015	0.019
S5/B1	2.295	0.002	0.000	0.000
S5/B2	-0.683	0.413	0.000	0.000
S5/B3	-1.240	0.222	0.000	0.000
S5/B4	-1.457	0.102	0.002	0.003
S5/B5	-1.678	0.196	0.079	0.092

Implications

- Ferson and Harvey (1999): even if you allow for time-varying betas, $\alpha_{i,t}$ is predictably changing
- Bad news for cost of capital measurement using the 3-factor model!
- Bad news for performance measurement using the 3-factor model!
- Bad news for everything based on the 3-factor model!

Wait a minute...

- If betas can move... what about Mr. CAPM?
- Can a time varying β_{CAPM} account for anomalies like size and value?
- Maybe it's all a matter of finding the correct z_t...

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Super important line of research during late 90s and early 2000s:

- Jagannathan and Wang (1996): betas conditional on past market return $+$ labor income risk can price the CS
- Lettau and Ludvigson (2001): conditioning betas on *cay* can price the CS
- Santos and Veronesi (2006): conditioning betas on labor income share can price the CS

Not So Fast, Guys

- Lewellen and Nagel (2006) killed this line of attack: the conditional CAPM cannot price the cross-section of returns
- Focus on using daily data instead of monthly returns
- Estimate betas for each month or quarter and collect alphas
- This avoids having to fully specify z_t

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	Size			B/M			Momentum		
	Small	Big	S–B	Growth	Value	V–G	Losers	Winners	W–L
Average conditional alpha (%)									
Quarterly	**0.42**	0.00	0.42	–0.01	**0.49**	**0.50**	**–0.79**	**0.55**	**1.33**
Semiannual 1	0.26	0.00	0.26	–0.08	**0.40**	**0.47**	**–0.61**	**0.39**	**0.99**
Semiannual 2	0.16	0.01	0.15	–0.12	**0.36**	**0.48**	**–0.83**	**0.53**	**1.37**
Annual	–0.06	0.08	–0.14	–0.20	0.32	**0.53**	**–0.56**	0.21	**0.77**

Questions?

Part IV: We Should Have Higher Standards...

The Factor-Zoo Explosion

During the 2000s and early 2010s:

- Many strategies with large expected returns and with no relation to betas appeared
- Examples: accruals, profitability, investment, asset growth, ...
- See Fama and French (2008) for an extensive analysis

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- Also: several papers proposing theories + factors that price the 25 portfolios

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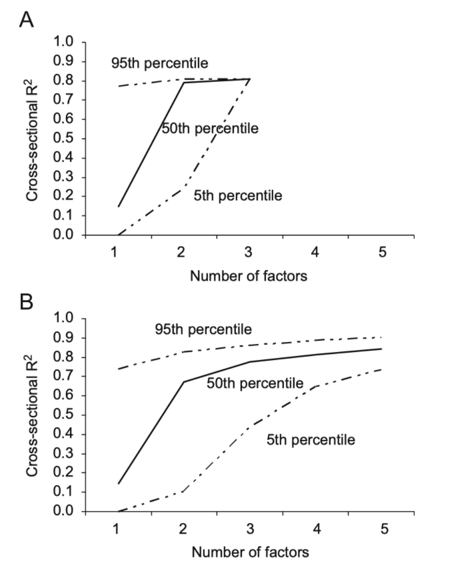
- Examples: accruals, profitability, investment, asset growth, ...
- See Fama and French (2008) for an extensive analysis
- Also: several papers proposing theories + factors that price the 25 portfolios
- Lewellen, Nagel, and Shanken (2010): this is too easy!

Lewellen-Nagel-Shanken: The Deeper Problem

- The 25 portfolios have a very strong factor structure
- We already know that MKT, SMB, and HML price them well
- If you give me a factor that is correlated with them… it will price the 25 portfolios too!
- We're judging models based on just a few assets!

A Killer Simulation



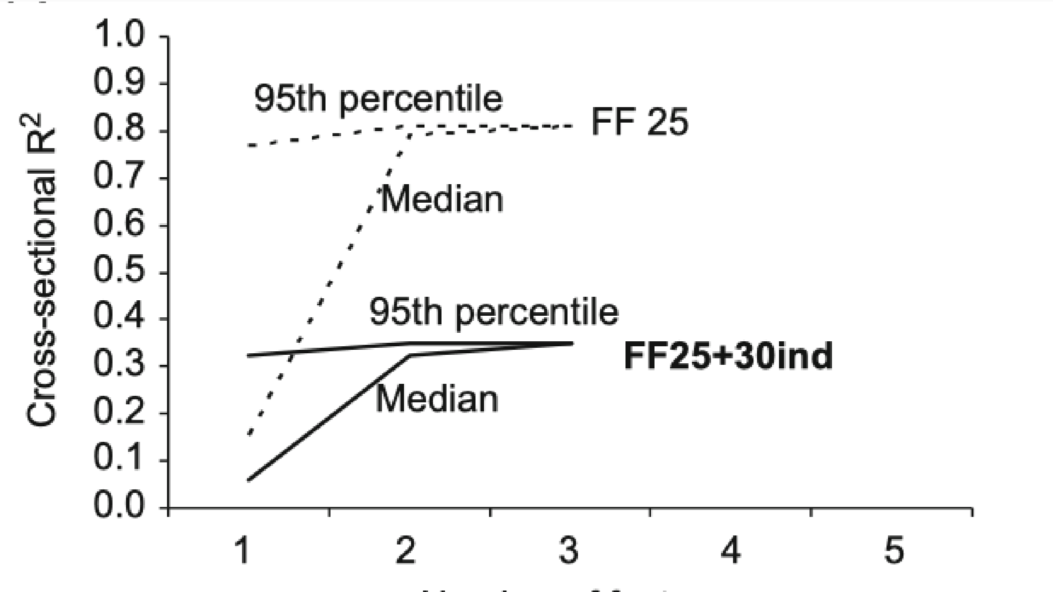
- Panel A: simulate random combinations of MKT, SMB, and HML
- Estimate the time-series betas and then run the cross-sectional pass
- Repeat many times and keep track of R^2
- Panel B: do the same, but with combinations of the 25 portfolios
- If you're not scared, look again

How To Improve Tests

They provide a few ways to improve the tests:

1. Expand the set of test portfolios beyond size-B/M portfolios
2. Take the magnitude of the cross-sectional slopes seriously
3. Report the GLS R^2 instead of OLS R^2
4. If a factor is traded, it should price itself (i.e., have a significant risk premium)
5. Report confidence intervals for the cross-sectional R^2

One Example of Improvement



Questions?

Part V: Characteristics Are Covariances

Part V: Characteristics Are Covariances

In construction

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